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ChatGPT in Academia: University Students' Attitudes Towards the use of ChatGPT and Plagiarism

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Abstract
This mixed-methods study examines the integration of ChatGPT within academic discourse, focusing on its role in education. ChatGPT, an AI chatbot using the GPT model, offers significant benefits such as enhanced plagiarism detection and improved accuracy in academic work. However, it also presents ethical challenges related to academic integrity. The research explores factors influencing student adoption, including perceived usefulness, ease of use, overall attitude, and anxiety related to ChatGPT usage. Additionally, the study investigates how gender, educational context, and prior AI experience affect ChatGPT usage and perception among a sample of 607 Hungarian university students. It also examines the potential relationship between students' willingness to use ChatGPT and their willingness to plagiarize. The findings reveal a generally positive attitude towards ChatGPT but also highlight concerns about potential misuse. This study aims to contribute to the responsible integration of AI in education by providing insights and recommendations to balance technological innovation with the preservation of academic integrity.

Keywords Academic integrity · Artificial intelligence · ChatGPT · Ethical concerns · Plagiarism

Introduction

The advent of artificial intelligence (AI) has revolutionized various domains, and education is no exception (Ding et al., 2023; Jo, 2024; Lee et al., 2024a; Sun et al., 2024; Tlili et al., 2023). Among the AI tools that have received significant attention is ChatGPT. ChatGPT is a conversational AI platform that employs natural language processing to engage in

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human-like interactions (Dwivedi et al., 2023). The widespread availability of ChatGPT, particularly its advanced version, ChatGPT 4.0, underscores its potential to significantly impact academic practices (Meyer et al., 2024). While the integration of AI like ChatGPT in education offers numerous advantages, including enhanced plagiarism detection and improved accuracy in academic work (e.g., Arabyat et al., 2022), it also presents challenges to academic integrity (e.g. Macdonald et al., 2023). The potential for misuse and the ethical implications of AI-generated content in scholarly contexts have sparked a vigorous debate (Wardat et al., 2023). The rise of such tools necessitates a re-evaluation of existing plagiarism guidelines and the establishment of new norms to address the unique challenges posed by AI (Waltzer et al., 2024).

This paper aims to explore the increasing role and importance of ChatGPT in academic settings, focusing on its advantages, ethical challenges, and the factors influencing its adoption and utilization by students. A critical gap in existing research is the lack of comprehensive understanding of the potential relationship between academic integrity, student engagement and AI tools. By addressing this research niche, this study seeks to contribute to the discourse on the responsible integration of AI in education and the development of guidelines that balance technological innovation with the preservation of academic integrity.

Theoretical Background

This study explores the incorporation of ChatGPT in academic discourse. It starts with a brief overview of ChatGPT and then discusses its potential benefits in education, such as enhancing academic integrity with plagiarism detection tools and improving the accuracy and efficiency of academic work. The study also examines factors that influence student adoption, including perceived usefulness, ease of use, overall attitude, and anxiety. Furthermore, it considers how usage and perception of ChatGPT vary based on gender, educational context, and prior AI experience. Finally, ethical concerns, particularly regarding academic integrity and potential plagiarism, are highlighted, emphasizing the need for a balanced approach that utilizes AI's capabilities while preserving human creativity and critical thinking.

ChatGPT and the Integrity of Academic Discourse

ChatGPT, an AI-driven chatbot developed by OpenAI, leverages the GPT model to produce human-like text responses (Dwivedi et al., 2023). This versatile tool finds applications in various fields, including answering queries (Liu et al., 2021), generating content (Lund & Wang, 2023), translating languages (Alawida et al., 2023), and coding (Kothari, 2023). Its extensive knowledge base, cultivated through processing vast datasets, empowers ChatGPT to deliver relevant information to users (Ming Li et al., 2024). A defining characteristic of ChatGPT lies in its capacity to engage in natural language interactions, allowing users to seek clarifications or modify response parameters (Garg et al., 2023). While the free ChatGPT 3.5 version demonstrates impressive text generation capabilities, the subscription-based ChatGPT 4.0 offers enhanced intelligence and precision (cf. Meyer et al., 2024). Despite the appeal of the free option, the superior performance of ChatGPT 4.0 might sway users towards the premium service.

Advantages of ChatGPT in Education

While concerns exist regarding its potential misuse, AI, exemplified by tools like ChatGPT, presents a double-edged sword for academic integrity (e.g., Dwivedi et al., 2023; Eke, 2023). On the one hand, AI technologies can contribute to enhancing academic integrity. They provide educators and institutions with advanced tools for improving plagiarism detection, which include AI-powered plagiarism-checking software like Turnitin's AI detection, as well as techniques developed by human researchers for textual analysis, metadata examination, and content evaluation (Willie, 2024) ensuring the integrity and authenticity of academic work (Arabyat et al., 2022; Gavilán et al., 2022). Additionally, AI can assist students in verifying the originality of their work and minimize citation errors by streamlining the referencing process (Jarrah et al., 2023). This technology serves as a valuable resource that contributes to enhancing the accuracy and efficiency of academic work, offering personalized support throughout individual learning journeys (Chan et al., 2023; Karataş et al., 2024; Tiwari et al., 2023; Zhou & Li, 2023).

Factors Influencing Student Adoption and Utilization of ChatGPT for Educational Purposes

Several studies have investigated how students adopt and utilize ChatGPT for educational purposes. These investigations focused on factors influencing student behaviour, such as *perceived usefulness*, which relates to the belief that the tool can significantly enhance learning outcomes by improving comprehension and efficiency (e.g., Abdaljaleel et al., 2024; Lee et al., 2024b). *Perceived ease of use*, which pertains to the simplicity and intuitiveness of interacting with the tool, was also investigated (e.g., Saif et al., 2023; Tiwari et al., 2023). Additionally, *overall attitude and willingness* towards using ChatGPT, which encompass the general perception of its benefits and effectiveness as an educational resource and influence the likelihood of its continued adoption, were examined (e.g., Chan et al., 2023; Zhou & Li, 2023).

One consistent finding across these studies was a positive student attitude towards ChatGPT when they viewed it as a valuable educational tool. This positive attitude translated into a substantial impact on their learning experience, as the investigated students perceived value in ChatGPT's ability to provide personalized support throughout their individual learning journeys (Abdaljaleel et al., 2024; Chan et al., 2023; Karataş et al., 2024; Saif et al., 2023; Tiwari et al., 2023; Zhou & Li, 2023).

Despite the potential benefits identified by some research, the issue of anxiety surrounding ChatGPT use cannot be entirely discounted (Abbas et al., 2024). While some research suggests a positive association between stress and ChatGPT adoption for managing deadlines, highlighting its perceived ease of use and effectiveness (Saif et al., 2023), the overall picture remains complex. Other studies reveal a potential influence of anxiety on technology adoption, with certain students exhibiting initial hesitation due to anxiety, although not necessarily hindering ultimate use (Budhathoki et al., 2024). Interestingly, research by Abdaljaleel et al. (2024) indicates low student anxiety regarding potential drawbacks like plagiarism or skill decline. This suggests a generally favourable perception of ChatGPT as a learning tool, despite lingering concerns about its reliability or impact on critical thinking skills.

Differences in ChatGPT Usage and Perception

There are several factors that significantly influence ChatGPT usage and perception, with notable differences in how individuals interact with the tool (e.g. Abbas et al., 2024; Damiano et al., 2024). As for gender differences, based on prior investigation, males reported higher perceived usefulness of ChatGPT for education, learning new concepts, ease of interaction, and flexibility compared to females. While females exhibited a higher frequency of ChatGPT usage, males spent more time per session. Additionally, females expressed greater concern about over-reliance on technology (Bouzar et al., 2024). Furthermore, gender moderates the relationship between pre-existing AI technology habits and actual ChatGPT use. The negative influence of established AI habits on actual use is stronger for females, suggesting a potential gender gap in technology adoption behaviour (Arthur et al., 2024).

Beyond gender, learning context also plays a crucial role in influencing user perceptions and adoption (Abbas et al., 2024; Damiano et al., 2024). Studies indicate higher usage among males, individuals with higher education levels, and experienced professionals (Lin et al., 2024). In addition, university full-time students, immersed in an academic setting focused on learning complex subjects, perceive ChatGPT as a valuable tool for knowledge acquisition. This contrasts with adult learners with work experience, who prioritize productivity and task management, utilizing ChatGPT for different purposes. Students often leverage ChatGPT for research and study assistance, highlighting how the perceived value of its functionalities varies based on the user's educational context (Jo, 2023).

Moreover, studies support a link between educational background and both attitudes towards and frequency of ChatGPT usage. Students generally view ChatGPT as a helpful resource for academic tasks, leading to a more positive perception and potentially more frequent use compared to adult learners with work experience (de Winter et al., 2024).

Additionally, prior experience with AI technology can influence how students engage with ChatGPT. Students with prior experience using AI tools are more likely to have stronger intentions to use ChatGPT and are also more likely to translate those intentions into actual use. This suggests that students' existing comfort levels with technology play a role in shaping their adoption of ChatGPT for learning purposes (Arthur et al., 2024). Apart from that, the learning environment can influence the utilization of such tools. Students experiencing substantial academic pressures can drive the adoption of AI tools (Abbas et al., 2024).

Perceptual and Affective Factors in Student Readiness for ChatGPT

Prior investigations identified several key factors influencing student adoption of ChatGPT, highlighting the interplay of perceptual and affective factors in technology adoption. (e.g. Chan et al., 2023; Lee et al., 2024b; Saif et al., 2023). Positive correlation was identified between students' familiarity with GenAI technologies, measured by their knowledge and frequency of use, and their willingness to adopt them in the future (Chan et al., 2023). It was also found that students who perceive ChatGPT as valuable for tasks such as summarizing complex topics, generating creative writing prompts, or practicing foreign languages demonstrate a higher likelihood of regular adoption and use. A user-friendly interface and clear instructions further streamline integration into learning routines (Lee et al., 2024b). Addi-

tionally, a positive perception of ChatGPT's credibility was identified that fosters favourable attitudes towards its educational applications (Tiwari et al., 2023).

Further investigation indicated that the use of ChatGPT can alleviate feelings of tension and pressure, while simultaneously enhancing interest and enjoyment in learning. Notably, students who perceived themselves as more competent in their learning experienced greater enjoyment, irrespective of their assessment of ChatGPT's value. (Zhou & Li, 2023). In addition to that, prior studies identified a correlation between low anxiety levels, low perceived risks associated with ChatGPT usage, and higher adoption and effective use in educational settings (Abdaljaleel et al., 2023; Saif et al., 2023). These findings highlight the complex nature of factors influencing student motivation, with ChatGPT potentially playing a role in both and perceived engagement (Zhou & Li, 2023).

Ethical Concerns and ChatGPT: The Challenge to Academic Integrity

Nevertheless, it is imperative to acknowledge that the integration of AI in education presents a contentious issue (e.g., Klyshbekova & Abbott, 2024). The growing reliance on text-generating tools like ChatGPT in academic pursuits raises concerns about scientific integrity (e.g., GPT-3, Thunström & Steingrimsson, 2022).

Academic dishonesty that directly violates the principles of scientific integrity, involves gaining an unfair advantage in academic work through unethical means, such as cheating or copying from others (Anderman & Murdock, 2011; Colnerud et al., 2009; Faucher & Caves, 2009). While academic dishonesty is generally understood, perceptions of its severity and frequency vary based on individual beliefs and the specific context (Balbuena & Lamela, 2015; Kamal et al., 2023; Maloshonok & Shmeleva, 2019; McClung & Schneider, 2015). Plagisarism, a specific type of academic dishonesty, occurs when someone presents another's work as their own without giving proper credit (Marques et al., 2019; McCabbe et al., 2001). This can be intentional or accidental. Intentional plagiarism involves a deliberate act of deceit, where an individual knowingly claims another person's intellectual efforts as their own. Accidental plagiarism can occur due to a lack of understanding or awareness about proper citation practices and academic standards (cf. Von Dran et al., 2001). In essence, plagiarism in higher education is the misappropriation of intellectual property by claiming someone else's ideas or writing as original (Carroll, 2007).

The ubiquitous availability of information online, coupled with the ease of sharing and manipulation through digital tools, has facilitated a concerning trend in plagiarism (Chiang, 2022; Islam et al., 2015; Sayed & Lento, 2015). The emergence of powerful AI tools like ChatGPT introduces a new layer of complexity to the discussion of academic integrity (e.g., Dillon et al., 2024; Gulumbe et al., 2024; Macdonald et al., 2023; Waltzer et al., 2024; Wardat et al., 2023). The classification of AI-generated content as plagiarism and appropriate citation practices remains subjects of debate (Cotton et al., 2023; Dillon et al., 2024; Koos & Wachsmann, 2023; Macdonald et al., 2023; Okaibedi, 2023; Wardat et al., 2023). The rapid advancement of language models necessitates a re-evaluation of existing plagiarism guidelines. Given the ethical complexities surrounding the use of these models in academic contexts, a critical examination of current regulations is imperative (Durak & Cankaya, 2024; Eke, 2023; Perkins, 2023; Waltzer et al., 2024).

To address these issues, several prestigious journals have implemented guidelines prohibiting the use of AI for text creation while permitting AI-assisted language refinement.

While AI offers undeniable advantages, it is crucial to recognize that human creativity, critical thinking, and problem-solving abilities remain indispensable (Grassini, 2023; Humble et al., 2024). A balanced approach that harnesses the power of AI while preserving human ingenuity is thus essential for maximizing its benefits in education and research (Esplugas, 2023; Urban et al., 2024; Xu et al., 2024).

Methods

This mixed-methods study, using a concurrent triangulation design, collecting both qualitative and quantitative data simultaneously (Creswell, 2022), aimed to explore participants' attitudes and perceptions concerning both ChatGPT and plagiarism in higher education. In line with the goals of the paper and the reviewed literature, the following questions were posed:

RQ1. What are Hungarian university students' attitudes towards ChatGPT?

RQ1.1 Do Hungarian university students' attitudes towards ChatGPT differ significantly between males and females?

RQ1.2 Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who aim for achieving good grades and those who do not?

RQ1.3 Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who want to meet their teachers' expectations and those who do not?

RQ1.4 Do Hungarian university students' attitudes towards ChatGPT differ significantly between scholarship holders and tuition fee-paying students?

RQ1.5 Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who work besides their studies and those who do not?

RQ2. What is the relationship between Hungarian university students' attitudes towards ChatGPT and their views on plagiarism?

Participants and Instruments

The research targeted Hungarian university students, employing convenience and snowball sampling techniques to gather 607 participants. All respondents were from the same university, which offers business programs, ensuring that the study could deeply investigate the students' attitudes and perceptions of plagiarism within a consistent academic environment, particularly within the context of business education, thereby reducing variability due to differing institutional policies or cultures. The sample comprised 41.7% males ($n=253$) and 58.3% females ($n=354$), reflecting a relatively balanced gender distribution that provides a

comprehensive view of the university's student population. The participants had an average age of 20.34 years (SD = 1.57), indicating a predominantly young adult demographic typical of undergraduate populations. This specific context was crucial for understanding how attitudes towards ChatGPT and plagiarism might be shaped by the academic pressures and cultural norms unique to this university setting.

For our research instrument, we used two validated questionnaires: one developed by Sallam et al. (2023) to assess our participants views on and attitudes to ChatGPT, and another by Howard et al. (2014) to measure participants' views on and attitudes to plagiarism (also in Fajt & Schiller, 2025). Together, these instruments comprised ten scales. Participants rated their agreement with these statements on a five-point Likert scale (1 = Strongly disagree to 5 = Strongly agree). The scales related to ChatGPT were as follows:

1. Perceived risk in general: 5 items (α = 0.770): measuring the overall risk students associate with using ChatGPT or similar technologies (e.g. AI).
2. Anxiety: 3 items (α = 0.743): gauging the anxiety levels students feel when using ChatGPT or similar technologies (e.g. AI).
3. Attitude to technology/Social influence: 5 items (α = 0.878): evaluating students' attitudes towards technology, influenced by social and peer opinions.
4. Perceived usefulness: 6 items (α = 0.851): assessing students' perceptions of ChatGPT's usefulness in their academic work and learning.
5. Behavioural/Cognitive factors: 3 items (α = 0.737): investigating the behavioural and cognitive aspects influencing students' use of ChatGPT, such as willingness to use ChatGPT.
6. Perceived ease of use: 2 items (α = 0.849): assessing how easy and user-friendly students find ChatGPT.
7. Perceived risk of use: 3 items (α = 0.742): measuring the specific risks students associate with using ChatGPT, like potential misuse or adverse consequences.

The scales related to plagiarism were as follows:

8. Factors exacerbating plagiarism: 8 items (α = 0.802), evaluating conditions that might lead to plagiarism.
9. Justification for plagiarism: 6 items (α = 0.682), examining the extent to which students justify engaging in plagiarism.
10. Severity and penalty: 8 items (α = 0.712), assessing views on the seriousness of plagiarism and the appropriateness of penalties.

The survey also collected demographic information and background variables such as gender, scholarship status, employment status, academic goals, and the significance of meeting teachers' expectations. To gain deeper insights, two open-ended questions were included to explore students' perceptions and experiences related to ChatGPT, aiming to uncover issues and personal experiences that closed-ended questions might miss.

Data Collection and data Analysis

Data for this study were gathered through an online survey conducted in Hungarian, targeting university students from mid-June to early July 2024. Participation in the study was entirely voluntary, and respondents were assured that their responses would remain confidential and completely anonymous. The study received approval from an ethics committee (Ethics approval number: 2024/06/02), and all procedures complied with the principles of the Declaration of Helsinki.

The data from the closed-ended questions were analysed using SPSS 28.0. Various statistical procedures were employed, including descriptive statistics to calculate mean scores (M) and standard deviations (SD) for each scale, thereby highlighting participants' general attitudes towards plagiarism and ChatGPT. Mean scores were categorized using cut-off points: scores below 2.50 were considered low, scores of 3.50 and above were high, and scores between 2.50 and 3.50 were moderate. Additionally, independent samples t-tests (at the $p<0.05$ level of statistical significance) were conducted to identify significant differences in perceptions based on demographic and background variables such as gender, scholarship status, employment status, academic aspirations, and the importance of meeting teachers' expectations. Pearson's correlation coefficients were also calculated to explore relationships among the scales measuring attitudes towards plagiarism and ChatGPT. Thematic analysis was used to examine the answers to the open-ended questions.

Results

This section begins with the presentation of mean scores and potential statistical differences among various subsamples. Following this, the relationships among the scales are examined. Finally, the section concludes with a report on the answers to the open-ended questions. The results are presented in the order of the research questions.

Descriptive Statistics and Reliability Measures

When answering the first research questions (RQ1 *What are Hungarian university students' attitudes towards ChatGPT?*), first, mean scores and standard deviations were calculated for the different scales (Table 1). Anxiety towards ChatGPT is moderate but varies widely

Table 1 Mean scores and standard deviations of scales

Scales	Mean	Standard Deviation
1. Perceived risk in general	3.27	0.88
2. Anxiety	3.11	1.06
3. Attitude to technology/social influence	3.18	1.01
4. Perceived usefulness	3.13	0.84
5. Behavioural/cognitive factors	2.68	1.06
6. Perceived ease of use	4.27	0.89
7. Perceived risk of use	3.31	0.92
8. Factors that exacerbate plagiarism	2.80	0.77
9. Justification for plagiarism	3.58	0.69
10. Severity and penalty of plagiarism	3.27	0.63

(M = 3.11, SD = 1.06), reflecting diverse levels of concern. Attitudes towards technology and social influence are also moderate (M = 3.18, SD = 1.01), with students finding ChatGPT moderately useful (M = 3.13, SD = 0.84) and easy to use (M = 4.27, SD).

There is some variability in how students perceive behavioural and cognitive factors, i.e., willingness to use ChatGPT (M = 2.68, SD = 1.06) and the risks of using ChatGPT (M = 3.31, SD = 0.92). The analysis indicates that Hungarian university students moderately recognize factors that lead to plagiarism (M = 2.80, SD = 0.77) and somewhat justify engaging in it (M = 3.58). They perceive the severity and penalties of plagiarism at a moderate level (M = 3.27, SD = 0.63) (cf. Fajt & Schiller, 2025) and have a similar perception of general risks associated with ChatGPT (M = 3.27, SD = 0.88).

Differences in Students' Perception of ChatGPT

The structure of this section follows the order of the sub research questions of RQ1, with each research question being addressed sequentially within the section. First, significant gender differences were investigated in the case of scales 4–6, which were answered by all 607 respondents (253 males, 354 females), and scales 7–10, which were answered by the 487 respondents who have used ChatGPT before (231 males, 256 females) (RQ1.1 *Do Hungarian university students' attitudes towards ChatGPT differ significantly between males and females?*). An independent samples t-test was conducted to compare perceived risk in general between males and females. There was a significant difference in scores for males (M = 3.00, SD = 0.88) and females (M = 3.45, SD = 0.83); t(605) = -6.41, $p < 0.001$, d = 0.85. This suggests that females perceive higher risk in general compared to males. An additional independent samples t-test was performed to compare anxiety levels between males and females. The results showed a significant difference in scores for males (M = 2.85, SD = 1.02) and females (M = 3.29, SD = 1.05); t(605) = -5.17, $p < 0.001$, d = 1.04. This suggests that females experience greater anxiety levels compared to males. Following this, an independent samples t-test was carried out to compare attitudes towards technology/social influence between males and females. There was a significant difference in scores for males (M = 3.53, SD = 0.93) and females (M = 2.94, SD = 1.00); t(605) = 7.45, $p < 0.001$, d = 0.97. This result shows that males have a more positive attitude towards technology and social influence than females. Subsequently, an independent samples t-test was performed to compare perceived usefulness between males and females. There was a significant difference in scores for males (M = 3.32, SD = 0.83) and females (M = 2.96, SD = 0.81); t(485) = 4.82, $p < 0.001$, d = 0.82. This suggests that males perceive ChatGPT as more useful compared to females. Next, another independent samples t-test was conducted to compare behavioural and cognitive factors influencing ChatGPT use between males and females. There was a significant difference in scores for males (M = 2.93, SD = 1.07) and females (M = 2.46, SD = 1.01); t(485) = 4.95, $p < .001$, d = 1.04. This indicates that males are more influenced by behavioural and cognitive factors compared to females. Then, an independent samples t-test was conducted to compare perceived use of ChatGPT between males and females. There was no significant difference in scores for males (M = 4.24, SD = 0.97) and females (M = 4.31, SD = 0.81); t(485) = -0.90, $p = 0.368$, d = 0.89. This suggests that there is no significant gender difference in perceived use of ChatGPT. Finally, an independent samples t-test was performed to examine the perceived risk of ChatGPT use between males and females. The results showed a significant difference in scores, with males (M = 3.12, SD = 0.93) scor-

ing lower than females (M = 3.47, SD = 0.87); t(485) = -4.33, $p < 0.001$, d = 0.90. This result indicates that females perceive a higher risk of using ChatGPT compared to males.

Next, regarding RQ1.2 (*Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who aim for achieving good grades and those who do not?*), it was investigated whether there were any significant differences between students who aim for achieving good grades and those who do not in scales 4–6, which were answered by all 607 respondents (519 aiming for good grades, 88 not aiming for good grades), and scales 7–10, which were answered by the 487 respondents who have used ChatGPT before (413 aiming for good grades, 74 not aiming for good grades). An independent samples t-test was run to compare perceived risk in general between students who aim for achieving good grades and those who do not. There was a significant difference in scores for students aiming for good grades (M = 3.32, SD = 0.87) and those who do not (M = 2.93, SD = 0.86); t(605) = 3.97, $p < 0.001$, d = 0.87. This suggests that students who aim for good grades perceive a higher risk in general compared to those who do not. In addition, an independent samples t-test was also carried out to compare anxiety levels between students who aim for achieving good grades and those who do not. There was a significant difference in scores for students aiming for good grades (M = 3.18, SD = 1.04) and those who do not (M = 2.67, SD = 1.08); t(605) = 4.21, $p < 0.001$, d = 1.05. This indicates that students who aim for good grades experience higher levels of anxiety compared to those who do not. Additionally, an independent samples t-test was performed to compare perceived risk of ChatGPT use between students who aim for achieving good grades and those who do not. There was a significant difference in scores for students aiming for good grades (M = 3.38, SD = 0.89) and those who do not (M = 2.91, SD = 0.99); t(485) = 4.05, $p < 0.001$, d = 0.90. This result indicates that students who aim for good grades perceive a higher risk of using ChatGPT compared to those who do not. For the remaining scales, no significant differences were detected between students who aim for achieving good grades and those who do not.

As a next step, regarding RQ1.3 (*Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who want to meet their teachers' expectations and those who do not?*), it was investigated whether there were any significant differences between students who want to meet their teachers' expectations and those who do not in scales 4–6, which were answered by all 607 respondents (441 wanting to meet expectations, 166 not wanting to meet expectations), and scales 7–10, which were answered by the 487 respondents who have used ChatGPT before (349 wanting to meet expectations, 138 not wanting to meet expectations). An independent samples t-test was conducted to compare perceived risk in general between students who want to meet their teachers' expectations and those who do not. There was a significant difference in scores for students who want to meet expectations (M = 3.33, SD = 0.88) and those who do not (M = 3.11, SD = 0.87); t(605) = 2.72, $p = 0.007$, d = 0.88. This suggests that students who want to meet their teachers' expectations perceive a higher risk in general compared to those who do not. Subsequently, an independent samples t-test was run to compare attitudes towards technology/social influence between students who want to meet their teachers' expectations and those who do not. There was a significant difference in scores for students who want to meet expectations (M = 3.14, SD = 1.03) and those who do not (M = 3.32, SD = 0.93); t(605) = -2.01, $p = 0.045$, d = 1.01. This result indicates that students who do not want to meet their teachers' expectations have a more positive attitude towards technology and social influence. An independent samples t-test was conducted to compare behavioural and cognitive

factors influencing ChatGPT use between students who want to meet their teachers' expectations and those who do not. There was a significant difference in scores for students who want to meet expectations (M = 2.61, SD = 1.07) and those who do not (M = 2.86, SD = 1.04); t(485) = -2.35, $p = 0.019$, d = 1.06. This indicates that students who do not want to meet their teachers' expectations are more influenced by behavioural and cognitive factors compared to those who do. No significant differences were found in anxiety, perceived usefulness, perceived use of ChatGPT, and perceived risk of using ChatGPT between the two groups.

As a final step, differences between scholarship holders and tuition fee-paying students were investigated using independent samples t-tests across various scales (RQ1.4 *Do Hungarian university students' attitudes towards ChatGPT differ significantly between scholarship holders and tuition fee-paying students?*). The results showed no significant differences between the two groups in perceived risk, anxiety, attitude to technology/social influence, perceived usefulness, behavioural/cognitive factors, and perceived risk of ChatGPT use. Similarly, differences between students who work besides their studies and those who do not were examined using independent samples t-tests across several scales (RQ1.5 *Do Hungarian university students' attitudes towards ChatGPT differ significantly between students who work besides their studies and those who do not?*). The findings indicated no significant differences between the groups in terms of perceived risk, anxiety, attitude towards technology/social influence, perceived usefulness, behavioural/cognitive factors, and perceived risk of ChatGPT use.

Correlations of Various Factors of Plagiarism and ChatGPT Attitudes

To understand the relationships between attitudes towards ChatGPT and various factors related to plagiarism among Hungarian university students (RQ2 *What is the relationship between Hungarian university students' attitudes towards ChatGPT and their views on plagiarism?*), Pearson's correlation analysis was conducted (Table 2). Only statistically significant correlations (at the $p < 0.05$ level) are reported and the focus of analysis here is mainly on attitudes towards ChatGPT.

Perceived risk in general has a weak positive correlation with perceived risk of using ChatGPT ($r = 0.82$) and anxiety ($r = 0.64$), suggesting that students who perceive higher general risks also see specific risks in using ChatGPT and feel more anxious about it. Anxiety

Table 2 Significant correlations among scales

	1	2	3	4	5	6	7	8	9	10
1. Perceived risk in general	1									
2. Anxiety	0.64	1								
3. Attitude to technology/social influence	−0.16		1							
4. Perceived usefulness	−0.22		0.61	1						
5. Behavioural/cognitive factors	−0.14		0.52	0.69	1					
6. Perceived use of use			0.13	0.19		1				
7. Perceived risk of use	0.82	0.67	−0.12	−0.12			1			
8. Factors that exacerbate plagiarism	0.13		0.32	0.32	0.32			1		
9. Justification for plagiarism			0.28	0.28	0.20	0.10		0.46	1	
10. Severity and penalty of plagiarism	−0.14	0.15	0.29	0.28	0.24		−0.16	0.54	0.69	1

also has a slight positive correlation with the severity and penalty of plagiarism ($r=0.15$), indicating that students who experience higher anxiety about technology also perceive plagiarism as more severe. Attitude towards technology and social influence shows notable positive correlations with perceived usefulness ($r=0.61$), perceived ease of use ($r=0.13$), and behavioural and cognitive factors ($r=0.52$). This suggests that students with positive attitudes towards technology are more likely to find ChatGPT useful, easy to use, and show more willingness to use it. Furthermore, perceived usefulness of ChatGPT correlates positively with behavioural and cognitive factors (willingness to use ChatGPT) ($r=.69$), indicating that students who find the technology useful are more likely to engage with it. The perceived ease of use shows a small positive correlation with perceived usefulness ($r=0.19$), indicating that ease of use slightly enhances the perceived utility of ChatGPT. However, perceived risk of use has slight negative correlations with attitude to technology ($r=-0.12$) and perceived usefulness ($r=-0.12$), suggesting that higher perceived risks may somewhat diminish positive attitudes towards technology and its perceived usefulness. Regarding plagiarism-related factors, factors that exacerbate plagiarism show positive correlations with perceived risk in general ($r=0.13$), attitude to technology and social influence ($r=0.32$), perceived usefulness ($r=0.32$), and behavioural and cognitive factors (i.e., willingness to use ChatGPT) ($r=0.32$). This indicates that students who recognize more factors leading to plagiarism also perceive higher general risks, have positive attitudes toward technology, and find ChatGPT useful. Justification for plagiarism correlates positively with attitude to technology and social influence ($r=0.28$), perceived usefulness ($r=0.28$), and behavioural and cognitive factors (i.e., willingness to use ChatGPT) ($r=0.20$). This suggests that students who justify plagiarism also tend to have favourable views of technology and find it useful. The severity and penalty of plagiarism show positive correlations with perceived risk in general ($r=-0.14$) and anxiety ($r=0.15$), as well as with factors that exacerbate plagiarism ($r=0.54$) and justification for plagiarism ($r=0.69$). These findings imply that students who perceive higher risks and feel more anxious about technology also view plagiarism as more severe and are more likely to acknowledge factors that exacerbate it and justify it.

Thematic Analysis of Student Perceptions on ChatGPT and Plagiarism

In addition to the quantitative survey instrument, a qualitative component employing open-ended questions was included in our investigation. This section allowed participants to elaborate on their perceptions of using AI (RQ1. *What are Hungarian university students' attitudes towards ChatGPT?*) and the perceived link between ChatGPT usage and plagiarism (RQ2. *What is the relationship between Hungarian university students' attitudes towards ChatGPT and their views on plagiarism?*), providing richer data beyond pre-defined answer choices. Although the sample size was limited to 30 voluntary participants, the qualitative data collected provided detailed insights (Fajt & Schiller, 2025), enabling a thorough analysis of student attitudes, concerns, and experiences related to AI in academic settings.

A thematic analysis (Braun & Clarke, 2020; Clarke & Braun, 2017; Maguire & Delahunt, 2017) was employed to analyse qualitative data and capture the participants' experiences in detail. Following Braun and Clarke's (2006) approach, a rigorous iterative coding process was implemented. First, we immersed ourselves in the data, generating initial codes. Second, we refined these codes into overarching themes. Finally, we critically reviewed these themes for accuracy and comprehensiveness. By systematically exploring the data, we

Table 3 Qualitative findings on participants' understanding and perceptions of ChatGPT usage and plagiarism

Core themes	Sub themes	Sample quotations (participants' ID number)
Acknowledging benefits while mitigating risks of AI	Ambivalence towards AI	"It can make our lives a lot easier, but there are two sides to everything. […] so inappropriate use may result in unforeseen consequences." (Participant no. 188)
	Limited trust in AI accuracy	"It is a misconception that it writes/solves everything for us because the answers are not always accurate." (Participant no. 375)
Strategies for responsible use	Strategic utilization of AI tools for source integration	In my opinion, it can help a lot if it is used normally. By that I mean not copying the answer straight from there, but just taking notes from it as a help and then incorporating it into your work." (Participant no. 87)
	Importance of verification and avoiding inaccuracy	"It can be useful, but it does not necessarily provide accurate information. I prefer to do more research either online or in the library or ask my peers." (Participant no. 193)
	Need for early education on responsible use of AI	"A very useful new advanced technology, but students should be taught at university as early as the first semester […] I think avoiding plagiarism is very important, and inappropriate use of ChatGPT leads to it." (Participant no. 226)

were able to investigate the depth and richness of the participants' experiences, uncovering valuable insights into their perspectives and motivations (Perena, 2023).The final coding constructs encompassing both core and subordinate concepts, were defined and illustrated with participant examples (Fajt & Schiller, 2025). Peer coding ensured the analysis' trustworthiness by fostering consistency and mitigating coder bias (Thomas & Magilvy, 2011).

The qualitative data surrounding ChatGPT usage revealed two key thematic units related to plagiarism concerning acknowledging benefits while mitigating risks of AI and strategies for responsible use (Table 3). The first thematic unit centres on students' negotiation of AI's potential. Several participants acknowledged its benefits but express ambivalence. Additionally, a subtheme emerged around limited trust in AI's accuracy, highlighting the need for critical evaluation alongside potential benefits. These findings suggest a nuanced student perspective, recognizing both AI's potential and the importance of responsible use. The second core theme focuses on student perspectives on how to leverage ChatGPT responsibly to avoid plagiarism. Several students highlight the importance of strategic AI use for source integration to avoid plagiarism by advocating for using AI-generated content as a source for notetaking not direct copying. Further, certain students underscore the crucial step of verifying information from ChatGPT to ensure accuracy and avoid using potentially false content. Finally, others acknowledge the risk of plagiarism but stress its importance to avoid by receiving proper guidance within a higher education context to mitigate it These strategies demonstrate student awareness of the potential pitfalls and their desire to use ChatGPT ethically within academic settings.

Overall, the qualitative analysis sheds light on student perceptions of AI use and their grasp of the potential for plagiarism associated with ChatGPT. While some students voice concerns about misinformation, others propose strategies for responsible use. These strategies emphasize critical evaluation of AI-generated content, leveraging it for knowledge acquisition in a responsible manner, and proper source citation to ensure academic integrity.

Discussion

The study provides valuable insights into the integration of ChatGPT in academic settings, highlighting both its benefits and ethical challenges. ChatGPT's capacity to generate human-quality text, as underscored by Dwivedi et al. (2023), positions it as a valuable asset for content creation (Kothari, 2023). The students in our study found ChatGPT moderately useful (M = 3.13) and easy to use (M = 4.27), which supports the notion that the tool can significantly enhance the accuracy and efficiency of academic work (Jarrah et al., 2023). However, the moderate anxiety levels (M = 3.11) and perceived risk (M = 3.31) indicate that concerns about misuse and reliability persist, which echoes the ethical challenges related to academic integrity highlighted by Eke (2023). The study found that perceived correlates with students' willingness to use ChatGPT. This aligns with the notion that AI tools like ChatGPT may bolster academic integrity by assisting in plagiarism detection (Arabyat et al., 2022) and verifying the originality of students' work (Gavilán et al., 2022). However, the problem is that quantitative methods provide insights only into general willingness to use it (Chan et al., 2023; Zhou & Li, 2023) but not what they use it for (e.g. Fajt & Schiller, 2025).

The study's exploration of factors influencing ChatGPT adoption revealed significant gender differences. Males reported higher perceived usefulness (M = 3.32) and positive attitudes towards technology (M = 3.53) compared to females. This supports previous findings that males tend to find ChatGPT more useful for educational purposes (Bouzar et al., 2024). However, females exhibited higher anxiety levels (M = 3.29) and perceived higher risks (M = 3.45) associated with ChatGPT use. These gender differences align with Arthur et al. (2024), who found that pre-existing AI technology habits influence actual ChatGPT use more strongly for females, suggesting a gender gap in technology adoption behaviour.

In addition, perceptual and affective factors significantly influenced students' willingness to use ChatGPT. Positive attitudes towards technology and social influence and factors exacerbating plagiarism correlate with ChatGPT usage. This supports the findings of Chan et al. (2023) and Lee et al. (2024), who emphasized the importance of a user-friendly interface and clear instructions in fostering favourable attitudes towards ChatGPT. Additionally, consistent with prior research results, low anxiety levels and perceived risks were associated with higher adoption rates (Abdaljaleel et al., 2023; Saif et al., 2023).

The qualitative component of the study revealed two main themes: students' acknowledgment of ChatGPT's benefits and their strategies for mitigating AI-related plagiarism risks. While participants recognized the tool's potential (cf. Bauzar, 2024; Jo, 2023), they emphasized the importance of critical evaluation, responsible use, and proper citation to maintain academic integrity and avoid misinformation (cf. Damiano et al., 2024). This suggests that further investigations are needed to explore how students are utilizing ChatGPT in their academic activities, Additionally, there is a need for educational programs to teach students how to use ChatGPT properly and ethically, ensuring that they leverage its capabilities to enhance learning while maintaining academic integrity (e.g., Ali et al., 2024).

Nevertheless, the integration of ChatGPT in education definitely raises ethical concerns about academic integrity, particularly in relation to plagiarism (Klyshbekova & Abbott, 2024). The study's findings indicate that students who use technology more often and specifically use ChatGPT, tend to plagiarize more frequently. Furthermore, students who perceive higher risks associated with AI also recognize more factors leading to plagiarism and view its consequences as more severe. This reflects the complexity of ethical considerations

in using AI-generated content (Dillon et al., 2024; Macdonald et al., 2023). Additionally, those inclined to plagiarize are more likely to use ChatGPT, as indicated by a systematic literature review by Jarrah et al. (2023), which highlights concerns raised by several studies (Anderson et al., 2023; Bom, 2023; Gao et al., 2023; Kumar, 2023; Mijwil et al., 2023; Perkins, 2023) about the potential for AI to facilitate unreferenced, plagiarized writings with serious ethical implications, such as issues of authorship. The need for balanced guidelines that harness AI's capabilities while preserving human creativity and critical thinking is crucial (Grassini, 2023; Urban et al., 2024).

Conclusion

To understand the factors influencing student adoption and perception of ChatGPT, this study examined students' attitudes, behaviours, and experiences related to the AI tool. Our findings indicate that students perceive ChatGPT as moderately useful and user-friendly, while also expressing concerns about its potential misuse and reliability. Perceived is closely related to willingness to use ChatGPT. While university students, especially those with prior AI experience, found ChatGPT beneficial for academic tasks, its integration into educational contexts necessitates careful consideration of ethical implications, particularly regarding plagiarism. To understand the factors influencing student adoption and perception of ChatGPT, this study examined students' attitudes, behaviours, and experiences related to the AI tool. The study found that students generally viewed ChatGPT as useful and easy to use, enhancing academic accuracy and efficiency. However, concerns about misuse and reliability persisted. Gender differences revealed that males perceived ChatGPT as more useful and had more positive attitudes towards technology, while females reported higher anxiety and perceived risks. Positive attitudes and social influence were significant predictors of ChatGPT usage, with low anxiety and perceived risks associated with higher adoption rates. Additionally, frequent technology users and those perceiving higher AI risks were more likely to recognize the severity of plagiarism's consequences.

As all research, this paper also has several limitations that need to be addressed. Firstly, the sample consisted exclusively of Hungarian university students from a single institution, which may limit the generalizability of the findings to other contexts and populations. Additionally, the reliance on self-reported data could lead to social desirability bias, where participants might overreport positive attitudes and behaviours or underreport negative ones (e.g., willingness to plagiarise) (Caputo, 2017). Another limitation is the cross-sectional design of the study itself, which captures students' attitudes and perceptions only at one specific moment (Bowen & Wiersema, 1999). A potential limitation of this mixed-method approach, which predominantly utilized quantitative investigation supplemented by open-ended questions, is its insufficient capacity to comprehensively capture the depth and intricacy of students' experiences and perceptions regarding ChatGPT usage. This limited qualitative component may constrain the exploration of nuanced insights and detailed practical challenges associated with AI integration in academic settings.

Future research should address these limitations in order to offer a more thorough understanding of ChatGPT's integration into academic settings. Expanding the sample to include students from diverse geographical regions and educational institutions would enhance the generalizability of the findings. In addition, longitudinal studies could track potential

changes in participants' attitudes over time, and mixed-methods approaches, combining quantitative surveys with in-depth qualitative interviews or focus groups, could offer a deeper exploration of how students integrate ChatGPT into their academic work. Furthermore, future research should investigate how ethical concerns surrounding ChatGPT use manifest across different educational contexts, such as varying institutional policies, fields of study, or levels of academic experience. This could provide insights into how specific academic environments shape students' perceptions and behaviours regarding AI tools. In addition, future researchers should focus on developing actionable strategies to address ethical challenges, such as creating clear guidelines for ChatGPT use in coursework or designing interventions to raise awareness of its ethical implications.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical Approval The study received approval from the ethics committee of the institution of the first author (Ethics approval number: 2024/06/02), and all procedures complied with the principles of the Declaration of Helsinki.

Competing Interests The authors declare that they have no competing interests that could bias the results of the manuscript.

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